

Module A (Lec 8): High-Dimensional Newton

Newton step (linear solver). Let $g_k = \nabla f(x^{(k)})$ and $F_k = \nabla^2 f(x^{(k)})$. Compute d_k from

$$F_k d_k = -g_k, \quad x^{(k+1)} = x^{(k)} + d_k.$$

- **Never form F_k^{-1} explicitly.** Use Cholesky (if $F_k \succ 0$) or iterative solvers (e.g. Conjugate Gradient).
- **Theorem 9.1 (Local Second Order Convergence)** Suppose that $f \in C^3$ and $x^* \in \mathbb{R}^n$ is a point such that $\nabla f(x^*) = 0$ and $F(x^*)$ is invertible. Then, for all $x^{(0)}$ sufficiently close to x^* , Newton's method is well-defined for all k and converges to x^* with an order of convergence at least 2. $\square$

Second-Order Optimality Conditions

- **SONC (Second-Order Necessary Condition):** If x^* is a local minimizer, then $\nabla f(x^*) = 0$ AND $\nabla^2 f(x^*) \succeq 0$ (PSD).
- **SOSC (Second-Order Sufficient Condition):** If $\nabla f(x^*) = 0$ AND $\nabla^2 f(x^*) \succ 0$ (PD), then x^* is a strict local minimizer.

Module A (lec 9): Levenberg–Marquardt (L–M) Modification

Theorem 9.2 (Newton Descent Direction) Let $\{x^{(k)}\}$ be the sequence generated by Newton's method for minimizing a given objective function $f(x)$. If the Hessian $F(x^{(k)}) \succ 0$ and $g^{(k)} = \nabla f(x^{(k)}) \neq 0$, then the search direction

$$d^{(k)} = -F(x^{(k)})^{-1}g^{(k)} = x^{(k+1)} - x^{(k)}$$

from $x^{(k)}$ to $x^{(k+1)}$ is a descent direction for f in the sense that there exists an $\bar{\alpha} > 0$ such that for all $\alpha \in (0, \bar{\alpha})$,

$$f(x^{(k)} + \alpha d^{(k)}) < f(x^{(k)}).$$

$\square$

Idea: enforce a descent direction even when $F_k \not\succ 0$.

$$x^{(k+1)} = x^{(k)} - (F_k + \mu_k I)^{-1}g_k, \quad \mu_k \geq 0.$$

- **Eigenvalue shift:** eigenvalues of $F_k + \mu_k I$ are $\lambda_i(F_k) + \mu_k$. Large enough μ_k makes it SPD.
- **Behavior:** $\mu_k \rightarrow 0 \Rightarrow$ Newton; $\mu_k \rightarrow \infty \Rightarrow$ small-step gradient.
- Often combine with line search:

$$x^{(k+1)} = x^{(k)} - \alpha_k (F_k + \mu_k I)^{-1}g_k.$$

Practice: Exercise 9.8 (Modified Newton + exact line search on a quadratic)

Consider

$$x^{(k+1)} = x^{(k)} - \alpha_k F(x^{(k)})^{-1}g^{(k)}, \quad \alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} - \alpha F(x^{(k)})^{-1}g^{(k)}).$$

Apply it to

$$f(x) = \frac{1}{2}x^\top Qx - x^\top b, \quad Q = Q^\top \succ 0.$$

Does it reach x^* in one step from any $x^{(0)}$?

Answer: Yes, it reaches x^* in exactly one step.

Proof: First, compute the Newton direction d_N . Since f is quadratic,

$$g(x) = Qx - b, \quad F(x) = Q.$$

The minimizer satisfies $Qx^* = b \implies x^* = Q^{-1}b$. The Newton direction is:

$$d_N = -F(x^{(0)})^{-1}g(x^{(0)}) = -Q^{-1}(Qx^{(0)} - b) = -(x^{(0)} - Q^{-1}b) = x^* - x^{(0)}.$$

Next, the exact line search seeks α_k to minimize $f(x^{(0)} + \alpha d_N)$. Notice that $x^{(0)} + \alpha d_N = x^{(0)} + \alpha(x^* - x^{(0)})$. This represents the line segment connecting $x^{(0)}$ and x^* . Since f is a strictly convex quadratic ($Q \succ 0$), the restriction of f to any line passing through the global

minimizer x^* is a 1D parabola with its unique minimum at x^* . Therefore, the minimizer along this direction occurs exactly when $x = x^*$, which corresponds to $\alpha = 1$. Since $\alpha_k = 1$ is chosen, $x^{(1)} = x^{(0)} + 1 \cdot (x^* - x^{(0)}) = x^*$.

Module B (Ch. 11): Quasi-Newton Methods — Preview

Form of the Family:

$$d_k = -H_k g_k, \quad x^{(k+1)} = x^{(k)} + \alpha_k d_k, \quad \alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} + \alpha d_k).$$

- H_k approximates F_k^{-1} using **only** function/gradient info (no Hessian factorization each step).
- **Descent condition:** $g_k^\top H_k g_k > 0$ (clean sufficient condition: $H_k \succ 0$).

Practice: Exercise 11.3 (Exact line search step size on a quadratic)

We run

$$d^{(k)} = -H_k g^{(k)}, \quad x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)}, \quad \alpha_k = \arg \min_{\alpha \geq 0} f(x^{(k)} + \alpha d^{(k)}),$$

with

$$f(x) = \frac{1}{2} x^\top Q x - x^\top b + c, \quad Q = Q^\top \succ 0.$$

- (a) Find a closed form for α_k in terms of Q , $g^{(k)}$, and $d^{(k)}$ (and hence H_k).

Answer: Let $\phi(\alpha) = f(x^{(k)} + \alpha d^{(k)})$. Then $\phi'(\alpha) = d^{(k)\top} (g^{(k)} + \alpha Q d^{(k)})$. So

$$\alpha^* = -\frac{g^{(k)\top} d^{(k)}}{d^{(k)\top} Q d^{(k)}}, \quad \alpha_k = \max \{0, \alpha^*\}.$$

If $d^{(k)} = -H_k g^{(k)}$ and $\alpha^* > 0$, then $\alpha^* = \frac{g^{(k)\top} H_k g^{(k)}}{(H_k g^{(k)})^\top Q (H_k g^{(k)})}$.

- (b) Give a sufficient condition on H_k for α_k to be positive (assume $g^{(k)} \neq 0$).

Answer: Since $Q \succ 0$, $d^{(k)\top} Q d^{(k)} > 0$ for $d^{(k)} \neq 0$. Thus $\alpha^* > 0 \Leftrightarrow -g^{(k)\top} d^{(k)} > 0 \Leftrightarrow g^{(k)\top} H_k g^{(k)} > 0$. A sufficient condition is $H_k \succ 0$.

Module C: Armijo Line Search (Backtracking)

Pick a descent direction d_k with $g_k^\top d_k < 0$ and choose α_k so that

$$f(x^{(k)} + \alpha_k d_k) \leq f(x^{(k)}) + c \alpha_k g_k^\top d_k, \quad c \in (0, 1).$$

- Start $\alpha = 1$; while Armijo fails, set $\alpha \leftarrow \beta \alpha$ ($\beta \in (0, 1)$, e.g. $\frac{1}{2}$).
- Terminates since $f(x^{(k)} + \alpha d_k) = f(x^{(k)}) + \alpha g_k^\top d_k + O(\alpha^2)$ and $g_k^\top d_k < 0$.